

```
from tensorflow.keras.datasets import fashion_mnist
```

```
(train_x, train_y), (test_x, test_y) = fashion_mnist.load_data()
```

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/train-labels-idx1-ubyte.gz
29515/29515 [=====] - 0s 0us/step
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/train-images-idx3-ubyte.gz
26421880/26421880 [=====] - 0s 0us/step
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/t10k-labels-idx1-ubyte.gz
5148/5148 [=====] - 0s 0us/step
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/t10k-images-idx3-ubyte.gz
4422102/4422102 [=====] - 0s 0us/step
```

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Flatten, MaxPooling2D, Conv2D
```

```
model = Sequential()
```

```
model.add(Conv2D(filters=64, kernel_size=(3,3), activation='relu', input_shape=(28, 28, 1)))
```

```
# Adding maxpooling layer to get max value within a matrix
model.add(MaxPooling2D(pool_size=(2,2)))
```

```
model.add(Flatten())
model.add(Dense(128, activation = "relu"))
model.add(Dense(10, activation = "softmax"))
```

```
model.summary()
```

```
Model: "sequential_2"
```

Layer (type)	Output Shape	Param #
flatten_1 (Flatten)	(None, 784)	0
dense_8 (Dense)	(None, 128)	100480
dense_9 (Dense)	(None, 10)	1290

=====

Total params: 101,770
Trainable params: 101,770
Non-trainable params: 0

```
model.compile(optimizer = 'adam', loss = 'sparse_categorical_crossentropy', metrics = ['accuracy'])
```

```
model.fit(train_x.astype(np.float32), train_y.astype(np.float32), epochs = 5, validation_split = 0.2)
```

```
Epoch 1/5
1500/1500 [=====] - 7s 5ms/step - loss: 0.4881 - accuracy: 0.8302 - val_loss: 0.5287 - val_accuracy
Epoch 2/5
1500/1500 [=====] - 6s 4ms/step - loss: 0.4893 - accuracy: 0.8298 - val_loss: 0.5376 - val_accuracy
Epoch 3/5
1500/1500 [=====] - 7s 5ms/step - loss: 0.4774 - accuracy: 0.8342 - val_loss: 0.5451 - val_accuracy
Epoch 4/5
1500/1500 [=====] - 6s 4ms/step - loss: 0.4751 - accuracy: 0.8361 - val_loss: 0.5717 - val_accuracy
Epoch 5/5
1500/1500 [=====] - 7s 5ms/step - loss: 0.4753 - accuracy: 0.8363 - val_loss: 0.5278 - val_accuracy
<keras.callbacks.History at 0x7fbcce0aceb0>
```

```
loss, acc = model.evaluate(test_x, test_y)
```

```
313/313 [=====] - 1s 2ms/step - loss: 0.5724 - accuracy: 0.8171
```

```
labels = ['t_shirt', 'trouser', 'pullover', 'dress', 'coat', 'sandal', 'shirt', 'sneaker', 'bag', 'ankle_boots']
```

```
predictions = model.predict(test_x[:1])
```

```
1/1 [=====] - 0s 73ms/step
```

```
import numpy as np
```

```
label = labels[np.argmax(predictions)]
```

```
import matplotlib.pyplot as plt
print(label)
plt.imshow(test_x[:1][0])
plt.show
```

```
ankle_boots
<function matplotlib.pyplot.show(close=None, block=None)>
```

